

Armandeep Singh

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SUMMARY

Backend developer specializing in scalable API architecture and real-time systems using Python and FastAPI. Built and deployed production-grade systems with authentication, rate limiting, and monitoring, focusing on performance, reliability, and clean system design.

SKILLS

- **Programming:** Python, JavaScript, SQL
 - **Backend & Web:** REST APIs, FastAPI, WebSockets, HTML, CSS, JWT
 - **Databases:** MySQL, SQLite, PostgreSQL, Database Design
 - **Tools:** Git, HTTPX, Docker, Redis
 - **Testing & Debugging:** API Testing, Debugging, Error Handling, Logging
 - **Soft Skills:** Problem Solving, Adaptability, Communication, Fast Learning
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PROJECTS

Pulse: Real-Time API Monitoring System ([live](#))

A production-ready backend system for real-time API health monitoring with live dashboards and secure multi-user access.

- Built a production-deployed real-time monitoring system (hosted on Render) tracking unlimited APIs with automated health checks every 30 seconds via configurable WebSockets.
- Implemented JWT-based authentication with bcrypt password hashing and secure cookie sessions.
- Designed REST APIs and a WebSocket-based live update system for instant status, latency, and uptime visualization.
- Developed a full admin panel (CRUD) with service grouping, validation, and dynamic updates.
- Integrated PostgreSQL (production) and SQLite (development) with environment-based configuration.
- Implemented structured logging, error handling, and scalable backend architecture.
- **Tech:** Python, FastAPI, WebSockets, SQLAlchemy, PostgreSQL, JWT, APScheduler, HTTPX

Accessor: API Key Management & Usage Tracking System ([live](#))

A production-deployed system mirroring the API key infrastructure used by companies like OpenAI and Stripe for controlling access, enforcing rate limits, and tracking usage in real time.

- Built a production-deployed API key management system with unique prefixed key generation (access...), JWT authentication, and bcrypt password hashing.
- Implemented Redis-based fixed-window rate limiting to enforce per-key request quotas with instant 429 Too Many Requests responses.
- Developed custom FastAPI middleware to intercept and log every request, tracking endpoint, latency, status code, and timestamp in PostgreSQL.
- Built a full analytics API providing per-key usage summaries including total requests, success/failure rates, and average response times.
- Designed and served a professional developer dashboard (Vanilla JS) with real-time usage streams and key management.
- **Tech:** Python, FastAPI, SQLAlchemy, PostgreSQL, Redis, JWT, Docker, Neon, Upstash, Render

INTERNSHIP & EXPERIENCE

06/2025 – 7/2025

Web Development Training - CGC Mohali

- Built modular Python scripts during a 1-month college training program, applying clean code principles and REST API integration.
 - Developed and optimized backend logic using Python, implementing efficient data handling and processing techniques.
 - Utilized Git for version control and integrated REST APIs to enable seamless data exchange and enhance application performance.
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EDUCATION

2023 - 2026	Bachelor of Computer Applications (BCA) (Present) CGC University, Jhanjeri, Punjab.	Current CGPA: 7.68
2022 - 2023	12th, PSEB Govt. Model Sen. Sec. School, Kharar, Punjab.	Percentage: 83%
2020 – 2021	10th, PSEB Vikram Public High School, Kharar, Punjab	Percentage:100%